MECHANICAL DESIGN & PRODUCT DEVELOPMENT PORTFOLIO

Gaurav Pandey
Master's Mechanical Engineering



WORKING AREAS

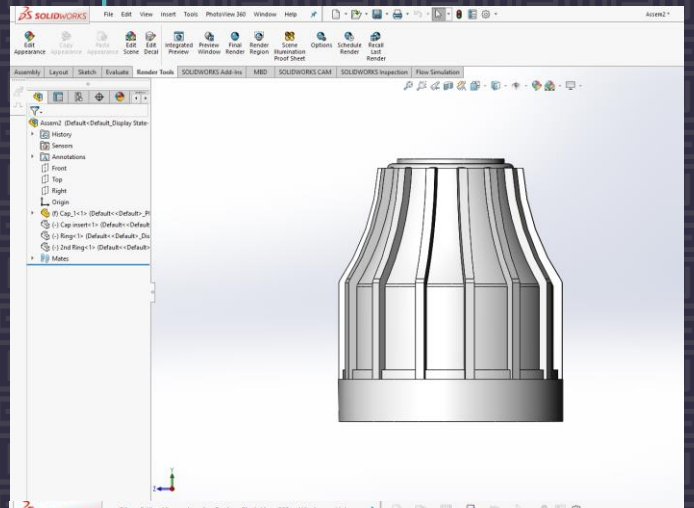
- Parts And Component Design/Reverse Engineering
- Tool Design
- Mechanism Design
- Product Development
- FEA Analysis

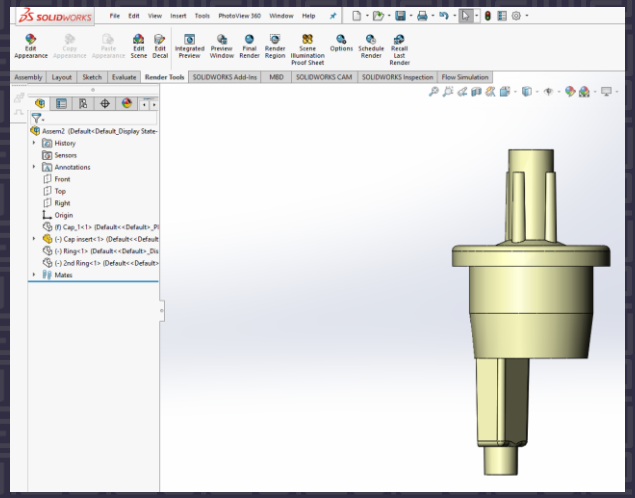
Note: Screenshots with software build interface history is displayed for Authenticity of design and software used

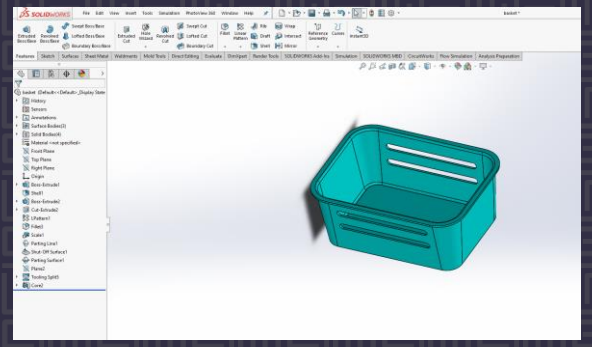
CAD AND CAE SOFTWARE USE

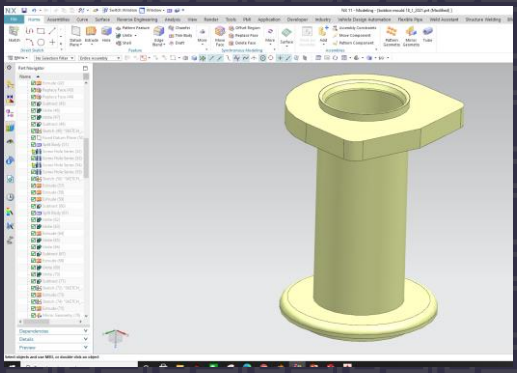
I. Solidworks 2016, 2018 & 2020
II. NX Unigraphics's Siemens 11.0
III. Ansys Workbench/Fluent 2019
IV. QForm Vx 8.2.3
V. Ebsilon
VI. Auto-CAD
VII. Blender

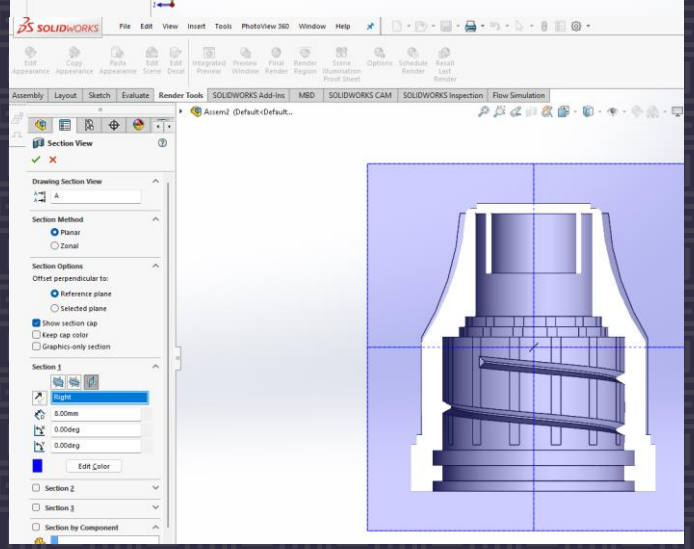
PART DESIGN [THERMOPLASTICS AND THERMOSETTING]

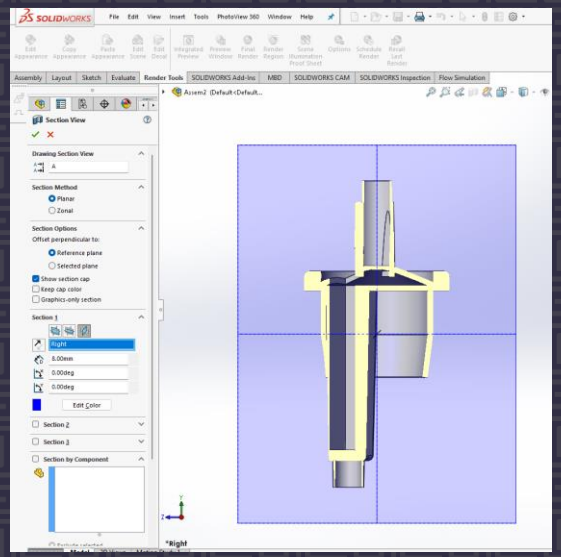


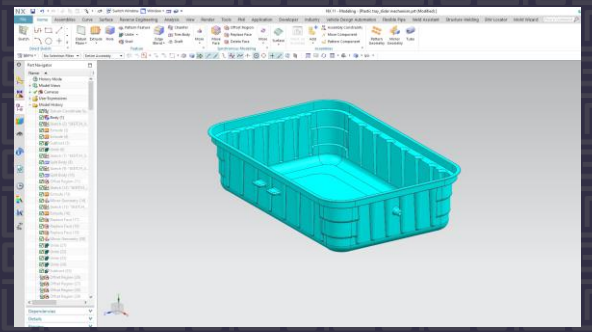


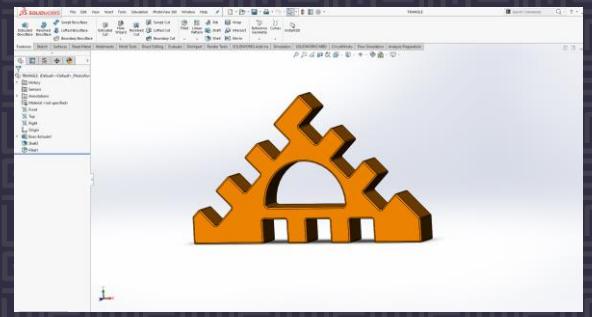






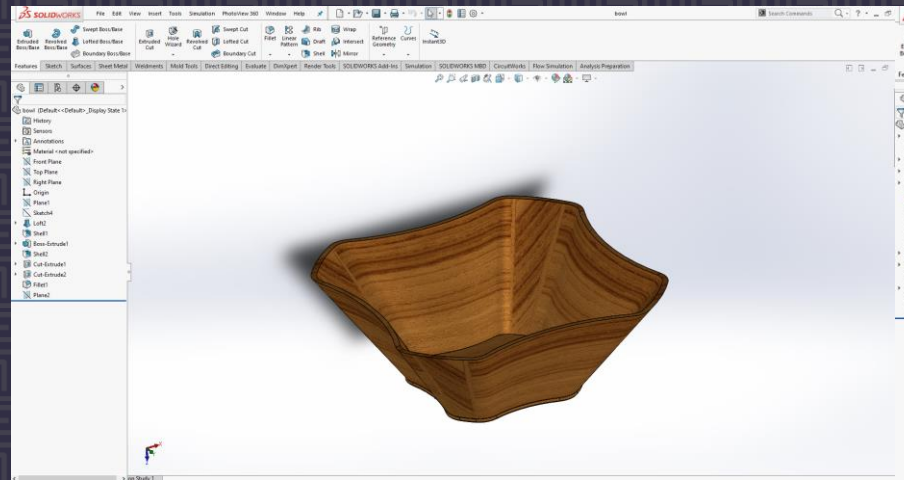


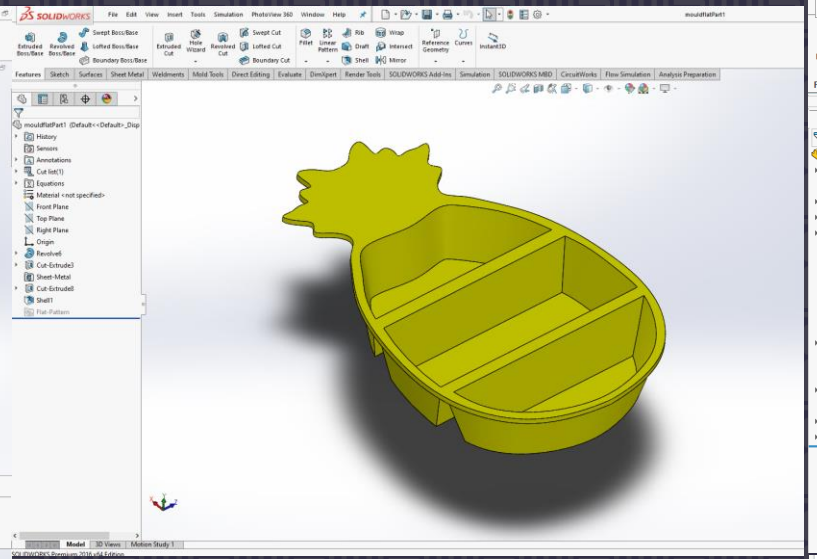


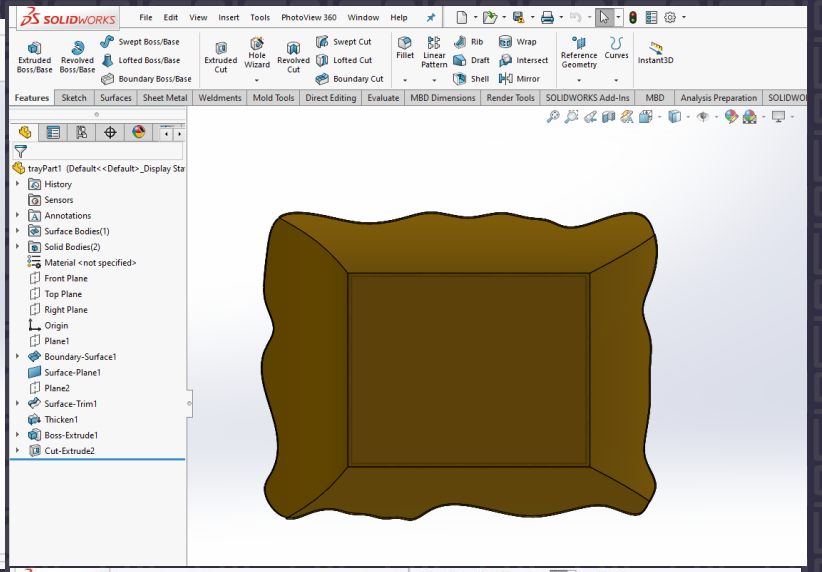


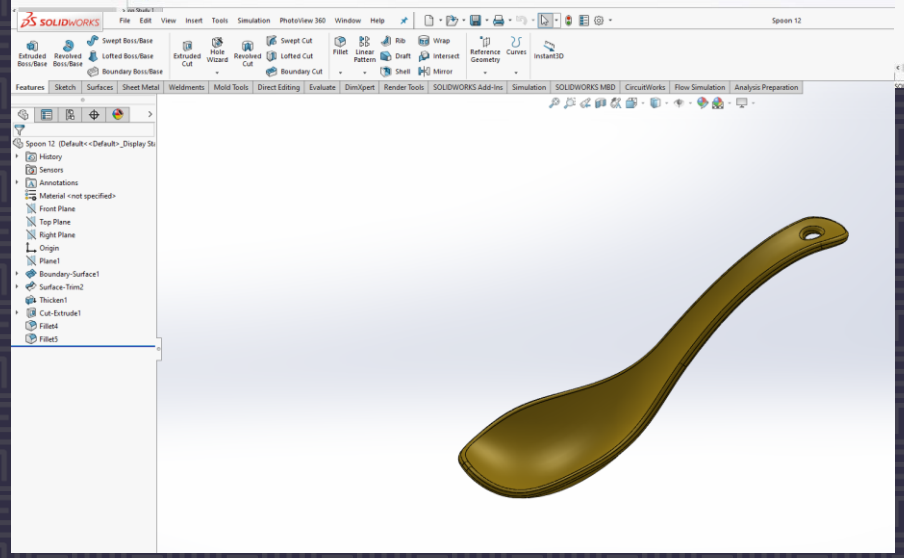
Thermoplastics-Medicine bottle Caps & Domestic Use parts [NX Unigraphics's &Solidworks]

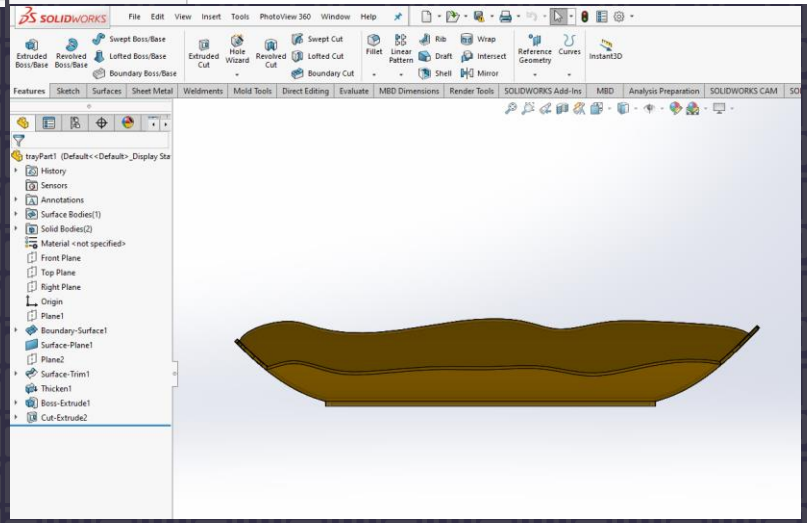
CONTINUE.....





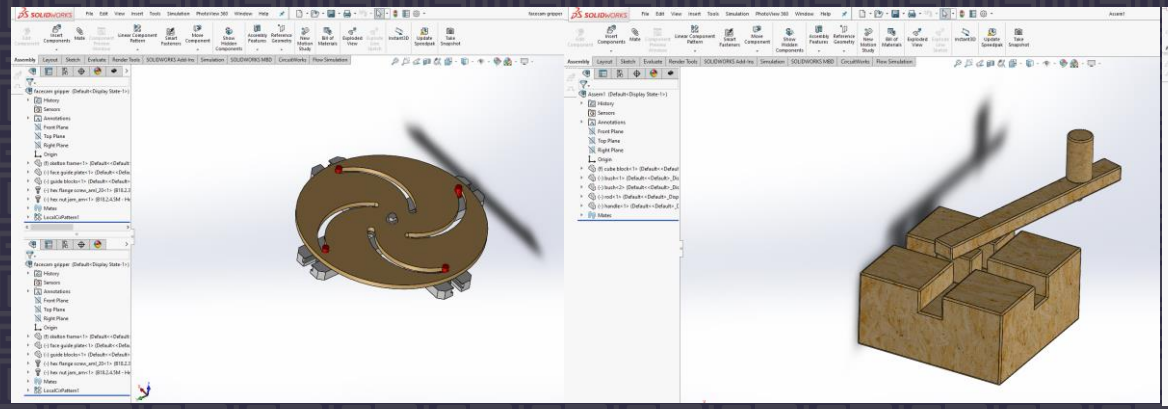






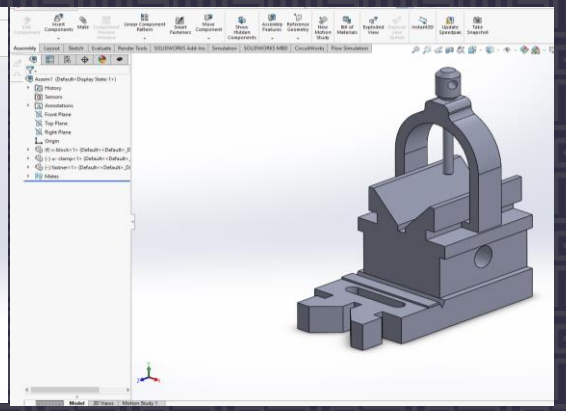
Thermosetting-Melamine Tableware Parts

MECHANISM DESIGN- ASSEMBLY

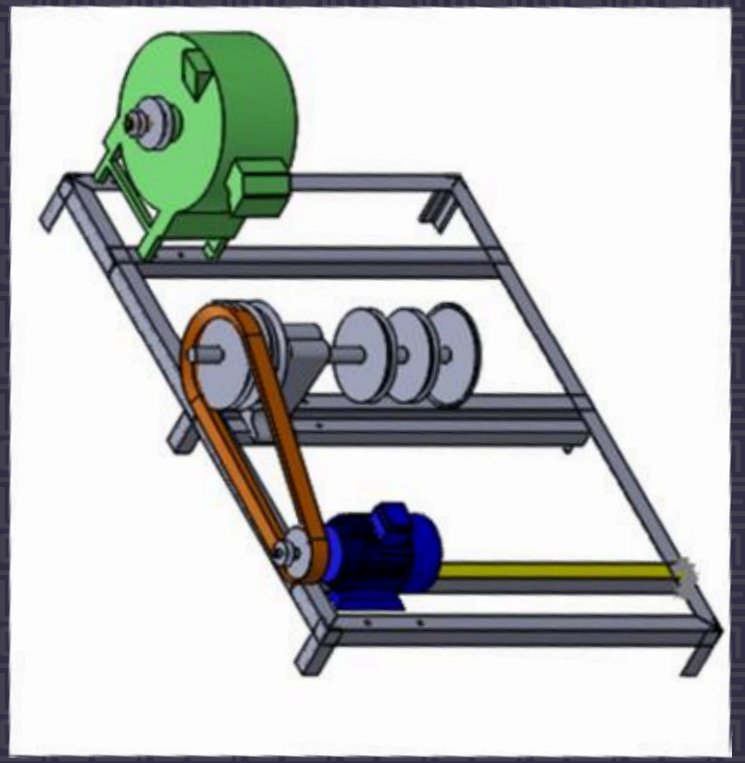


Mechanical Gripper

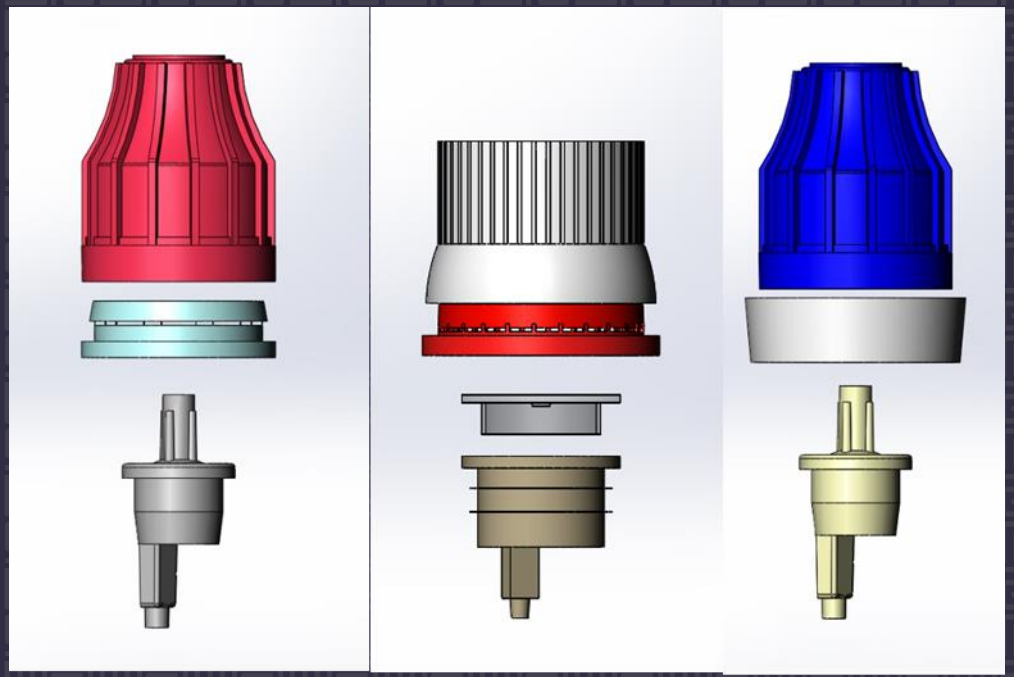
Toy



V-Block

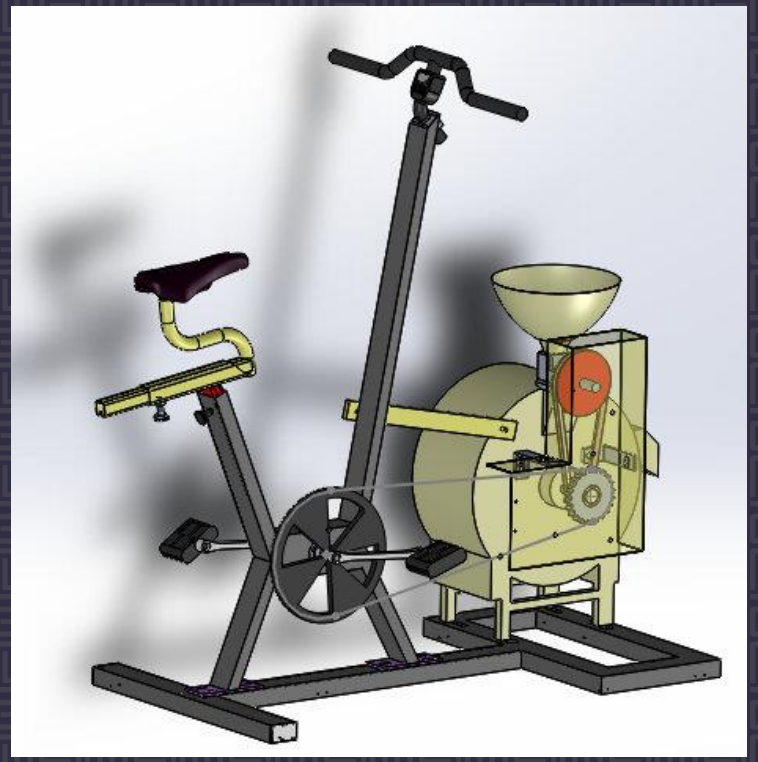


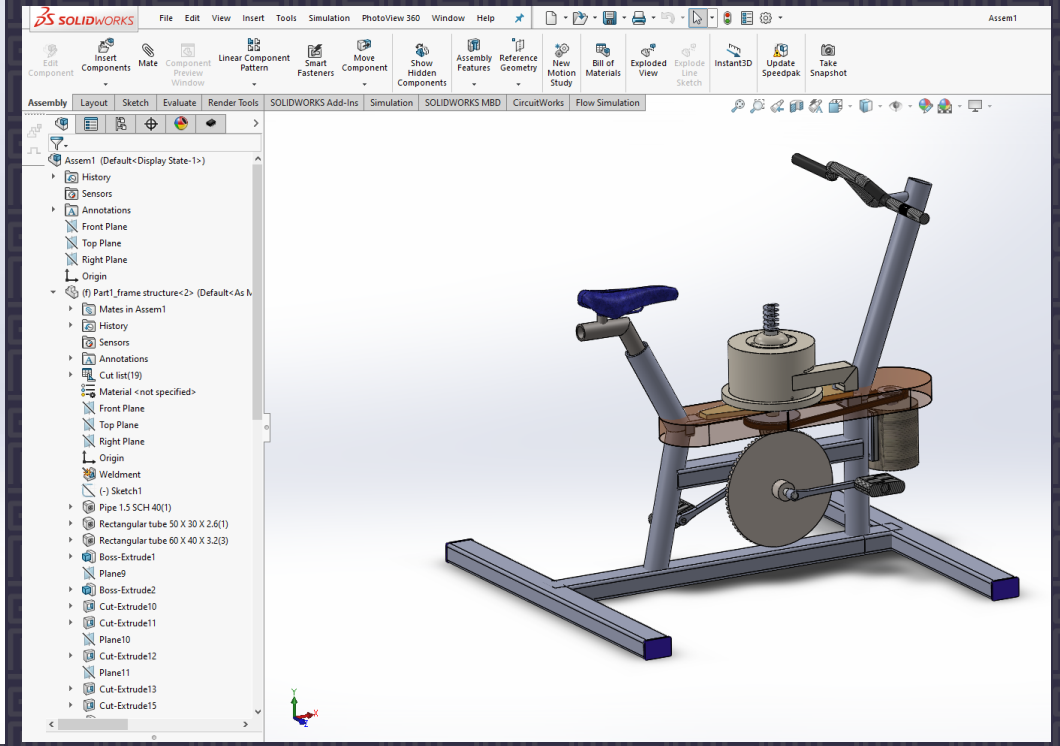
Testing Line for flour Chakki



Medical Cap-Screw unscrew with dropper

PRODUCT DEVELOPMENT
MULTI-FUNCTION CYCLE



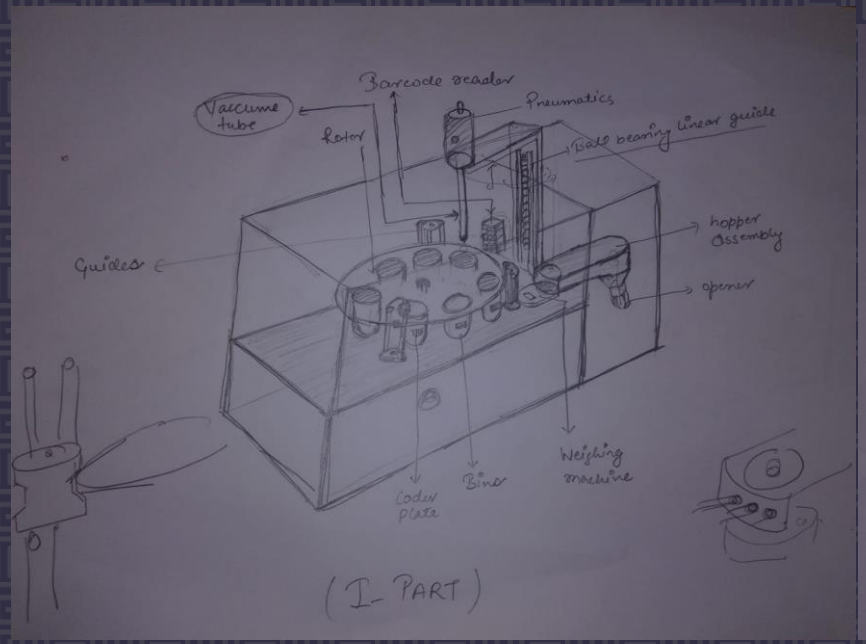


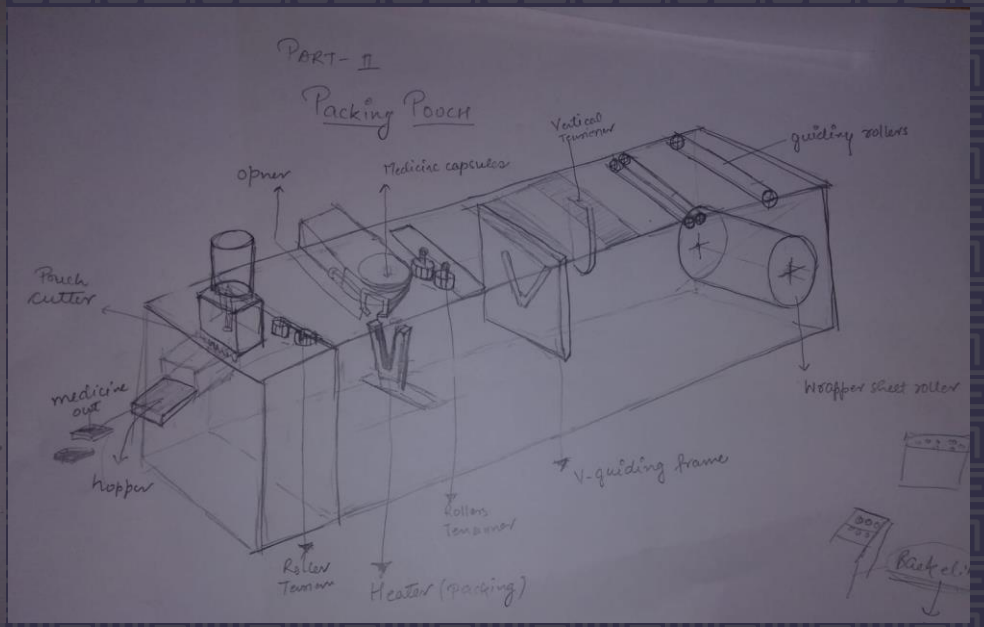


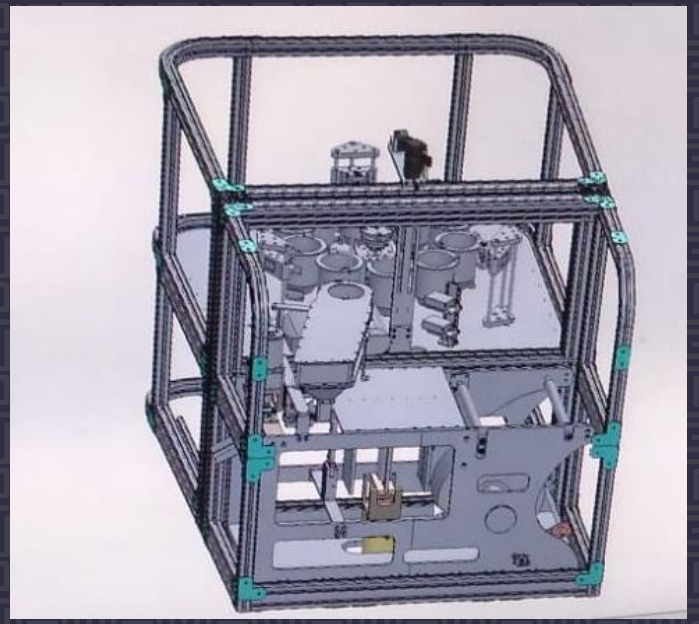
2017 → 2020

Advancement with weight reduction and part reduction along with Motor

AUTOMATED MEDICATION MACHINE

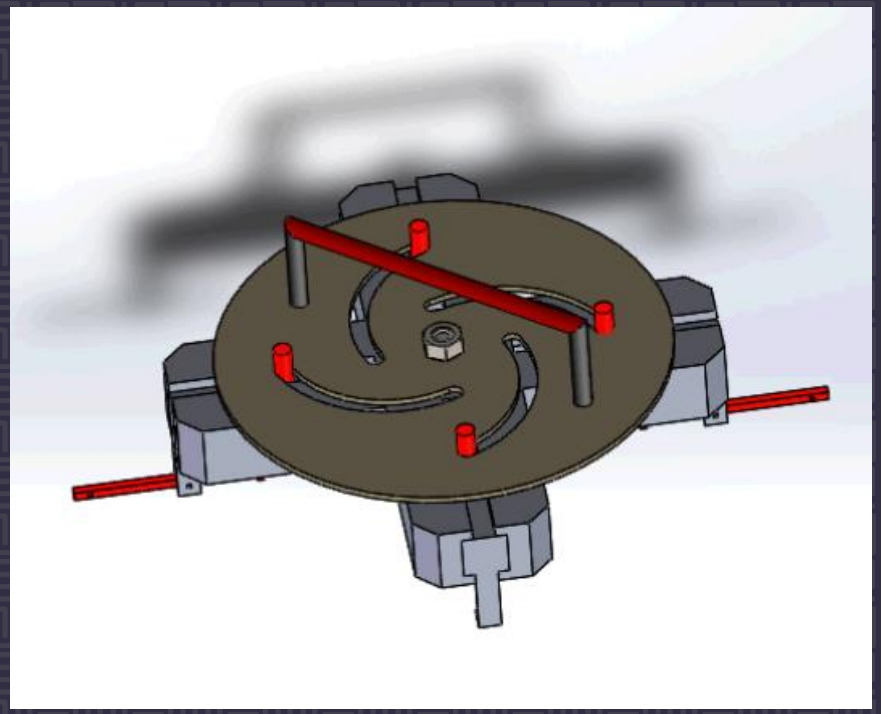


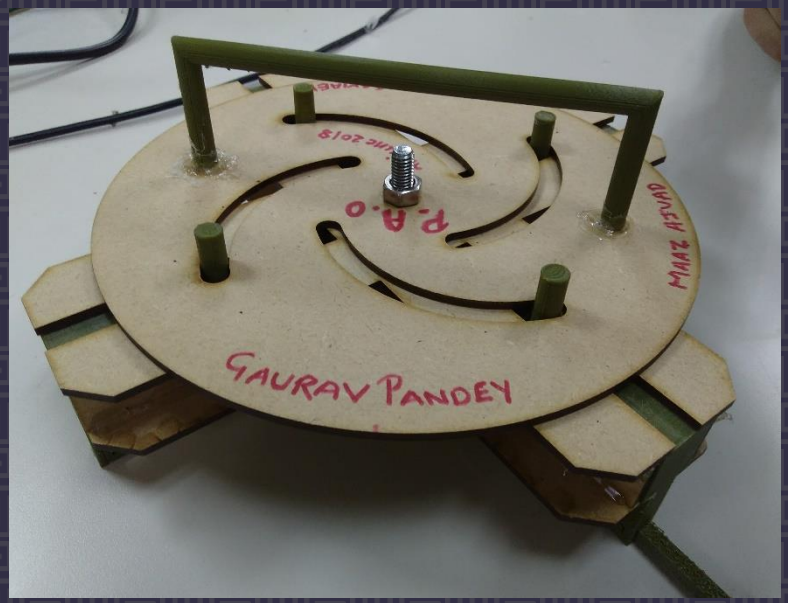




Design Layout plan from sketch to CAD modeling

MECHANICAL GRIPPER-PROTOTYPE DEVELOPMENT



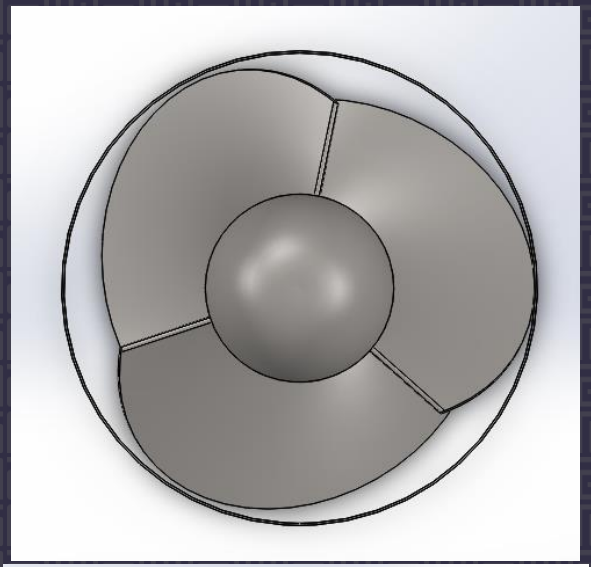


2017-19 Design Mechanism Project with rotary to linear mechanism capable of holding intrinsic shapes from light items like pin and eggs to heavy items like books under its projection range of structure.

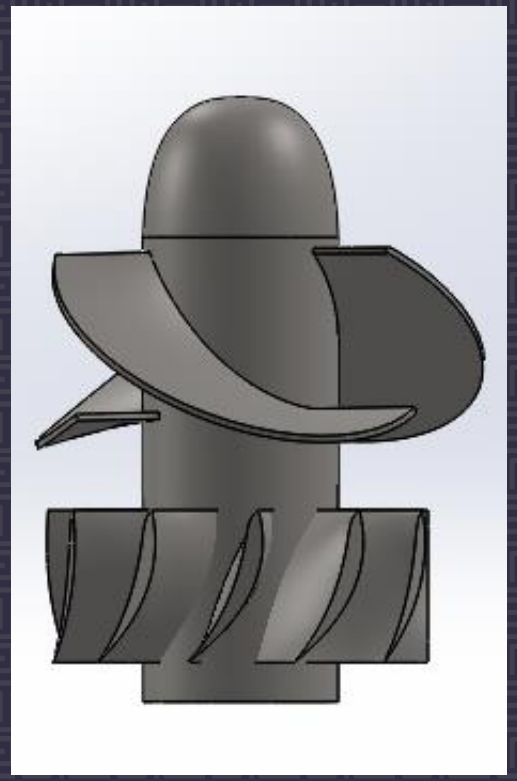
Manufacturing Process- 3-D printing and Laser cutting

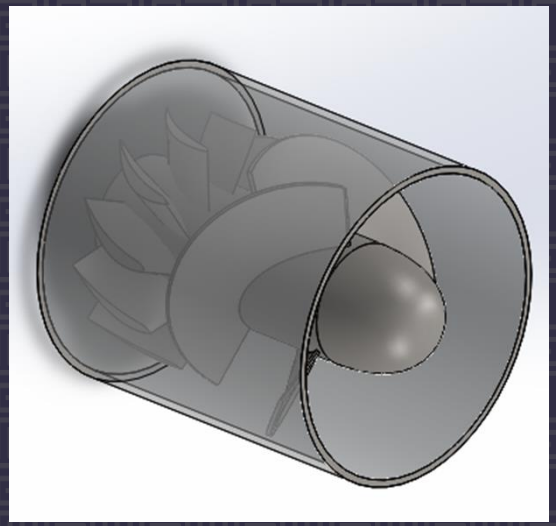
AXIAL PUMP DESIGN

s.no	variables	values
1	D	1.68 m
2	Dh	0.840 m
3	Dm	1.26 m
4	L	2m
5	No of impeller	3
6	No of diffuser	11
7	Casing	2.1 m

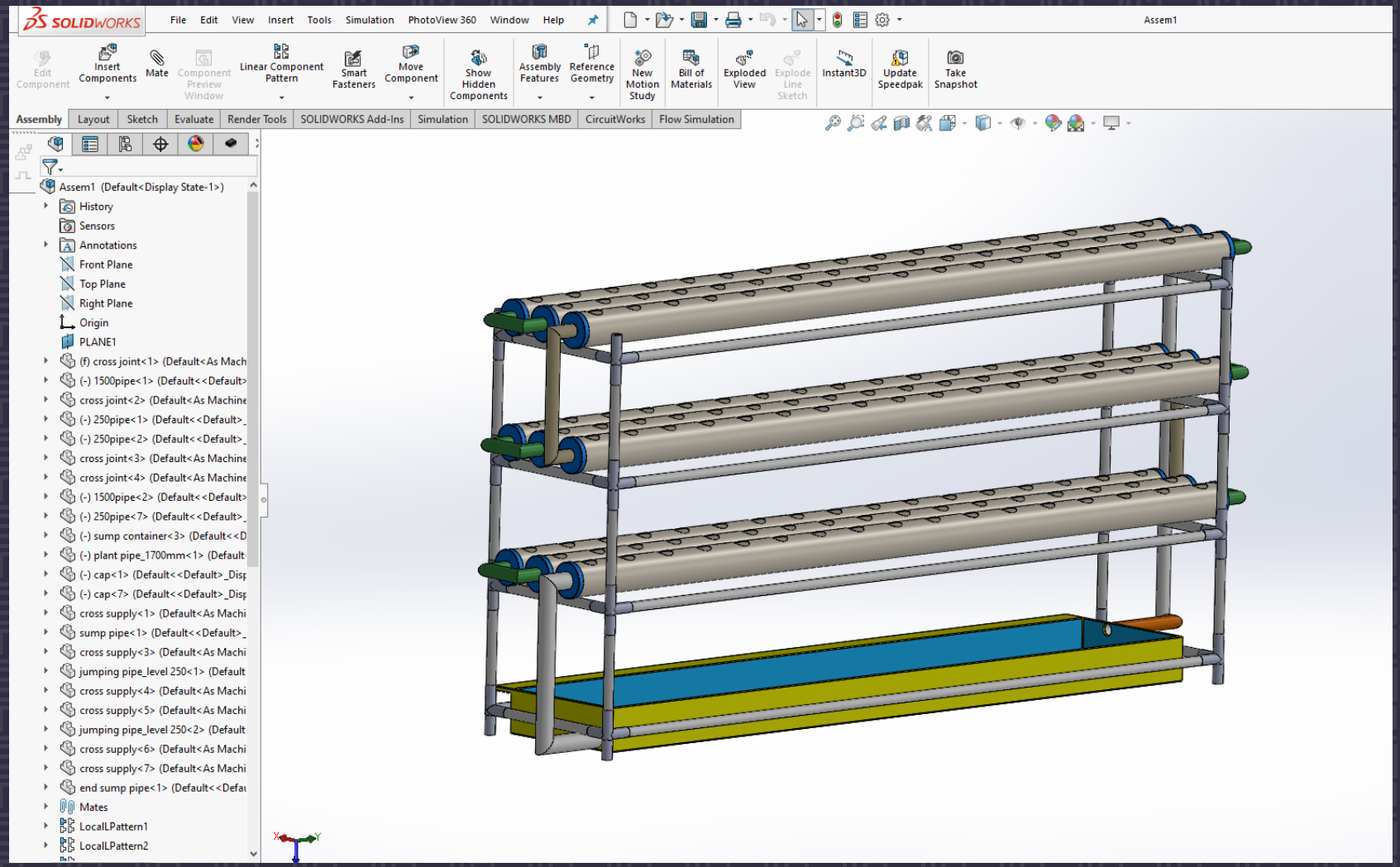






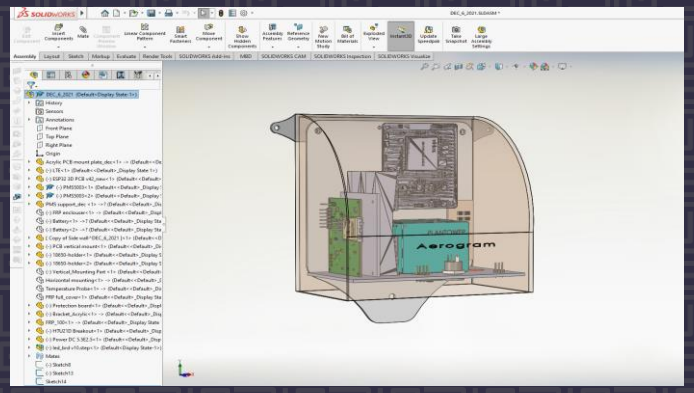


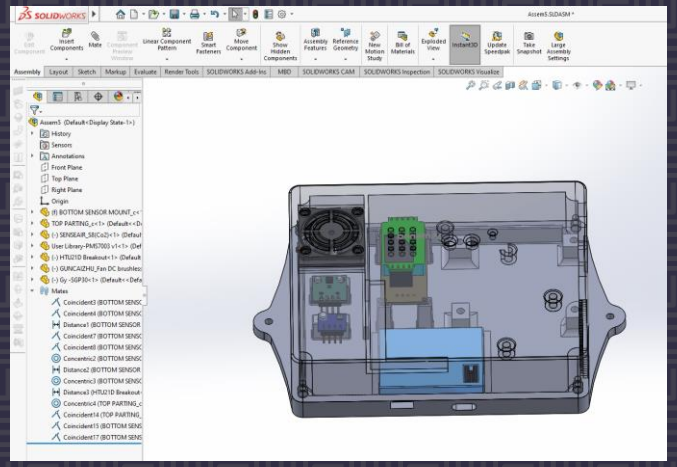
AUTOMATED HYDROPONIC SETUP

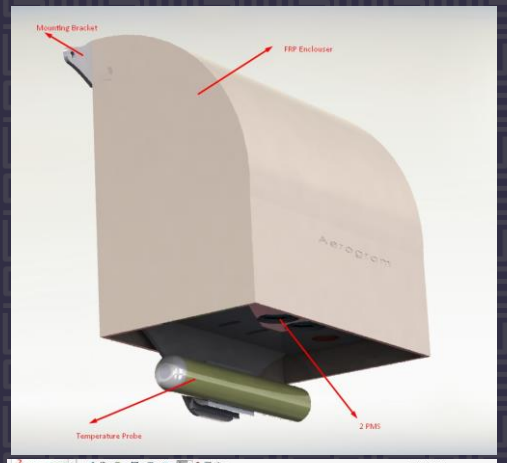


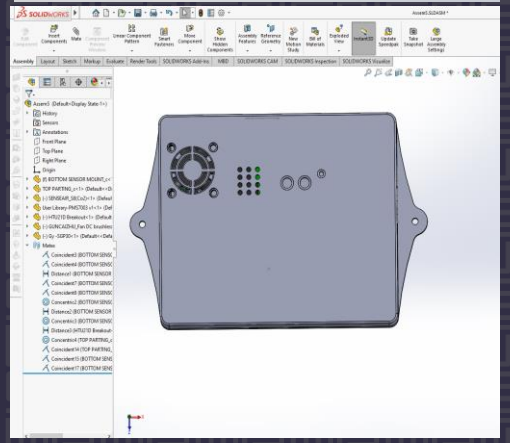
3-D plan design for hydroponic setup

MONITORING IOT DEVICES (AQI:PMS, TEMPERATURE, HUMIDITY, CO2) MEASURING DEVICE





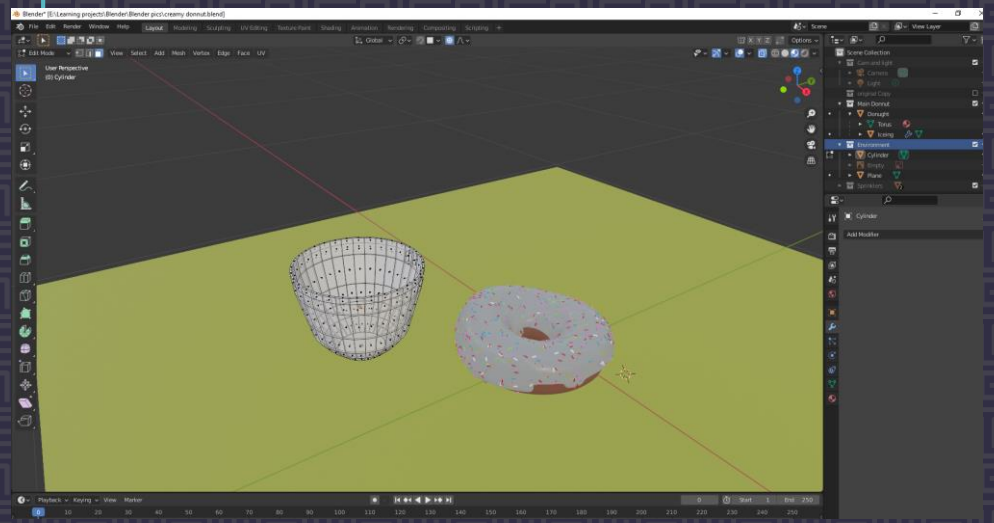




Product Names:
Ezio-Port (Outdoor), Ezio-stat (Indoor)

3-D Print, FRP enclosure with Electronics spread-out inside on acrylic base made via laser cutting, for AEROGRAM PVT Ltd.

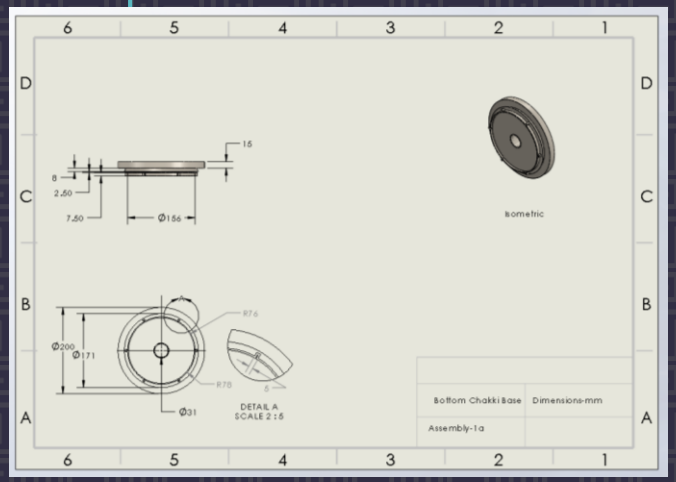
RENDERING AND DISPLAY

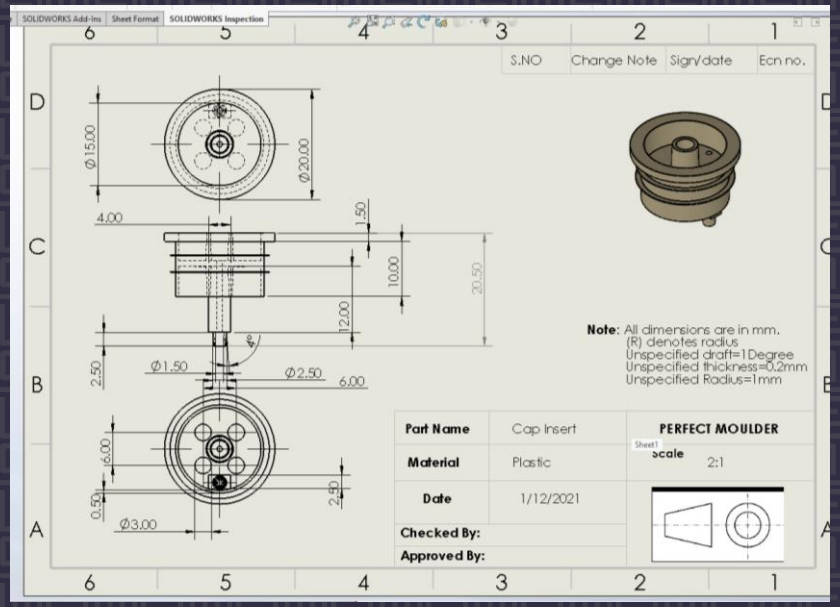


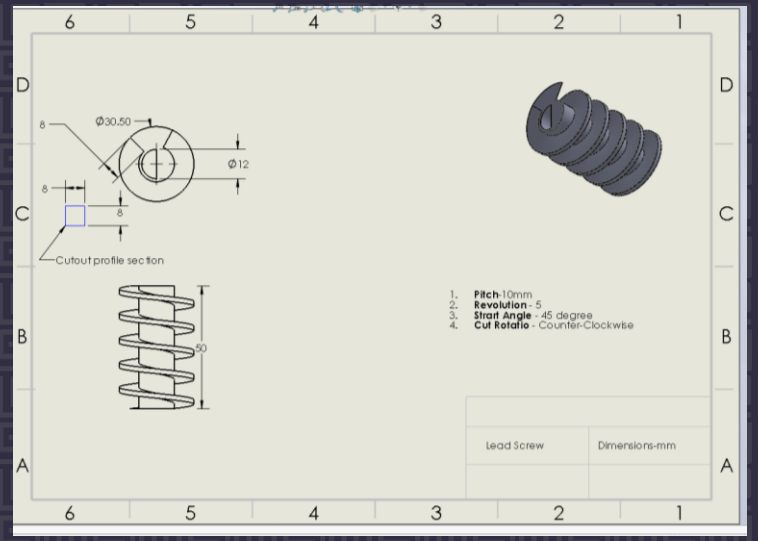


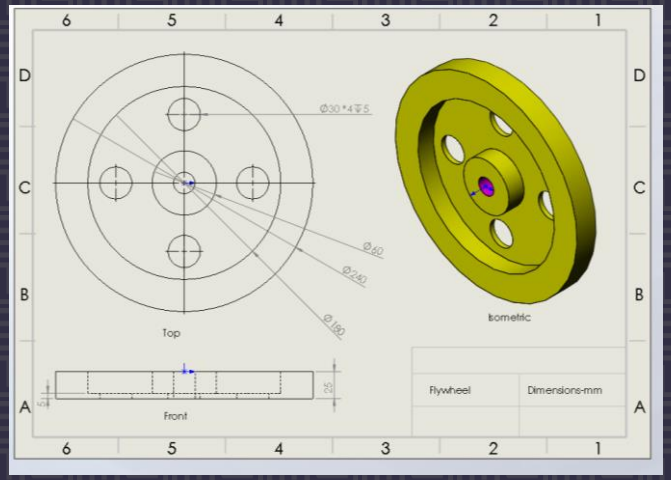


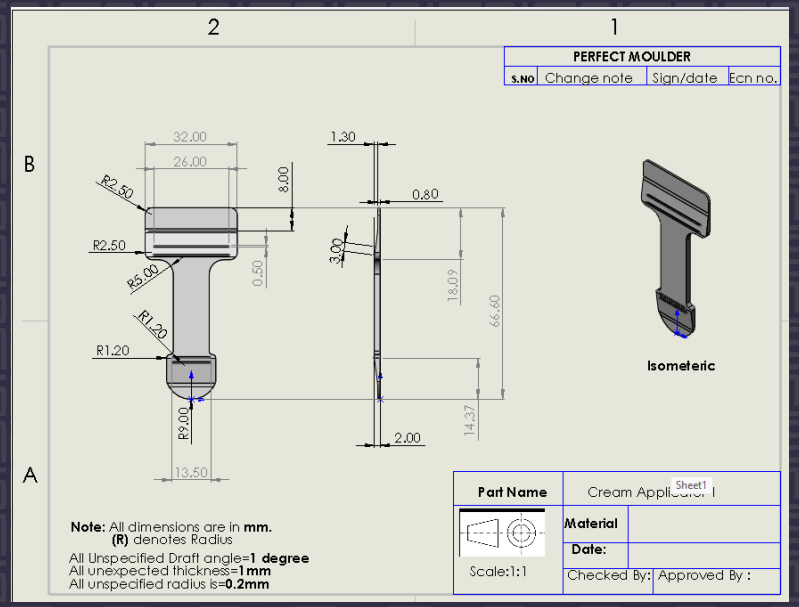
DRAFTING-1

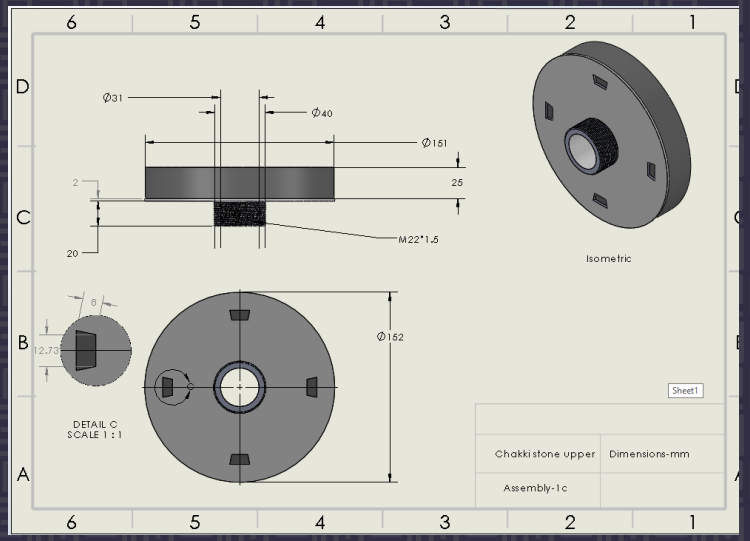






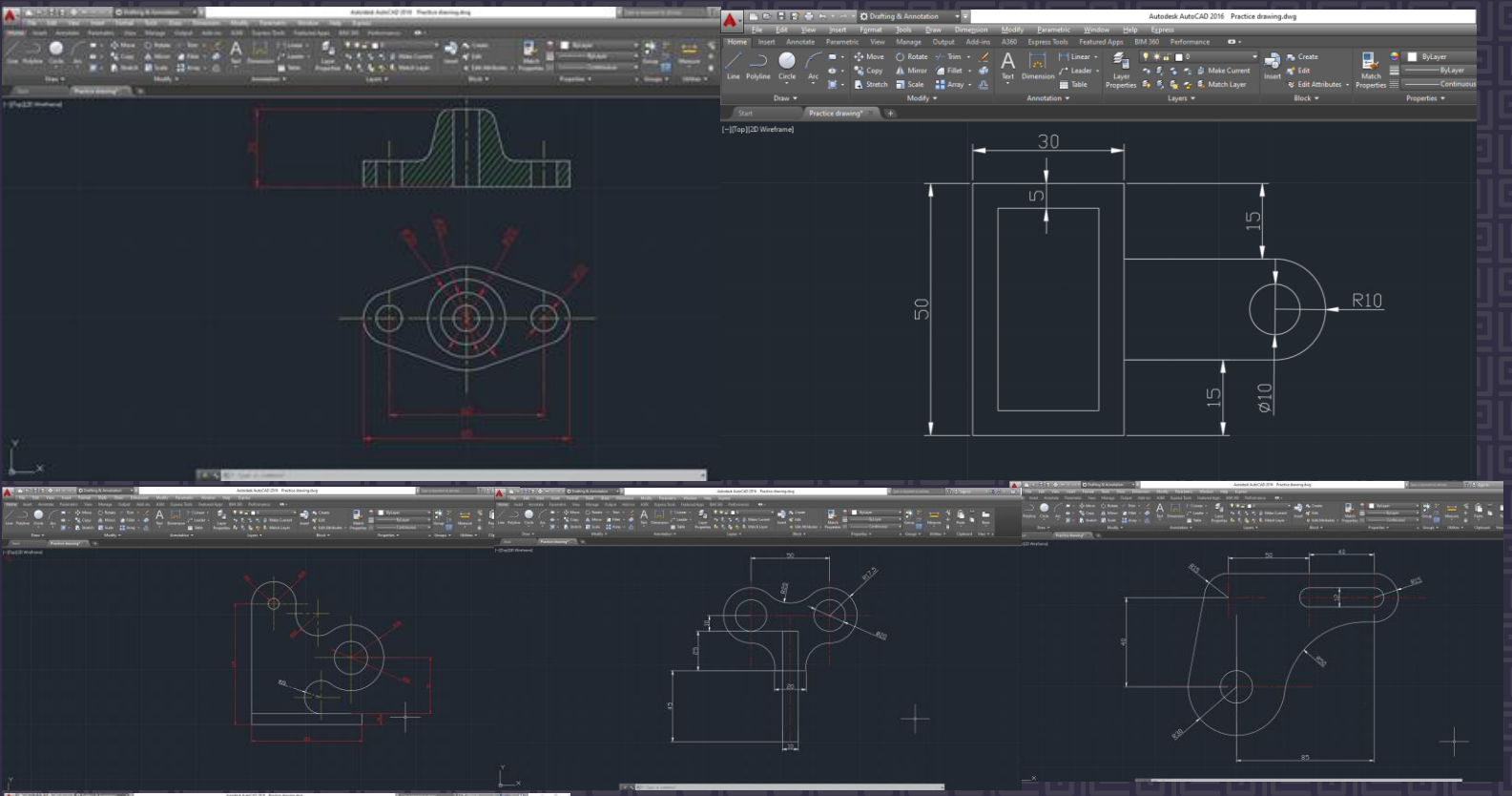




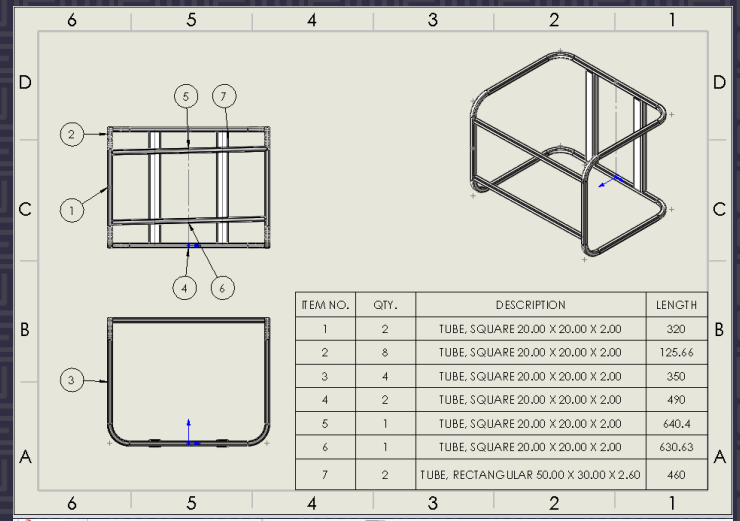


Solidworks drafting of 3-D parts

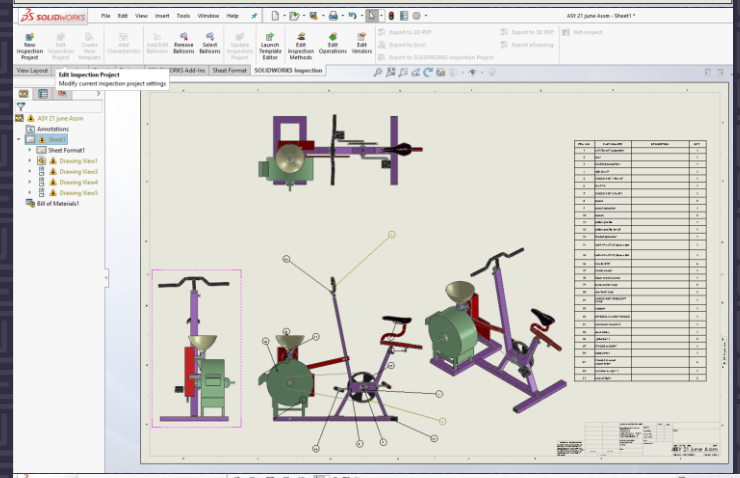
DRAFTING AND BOM

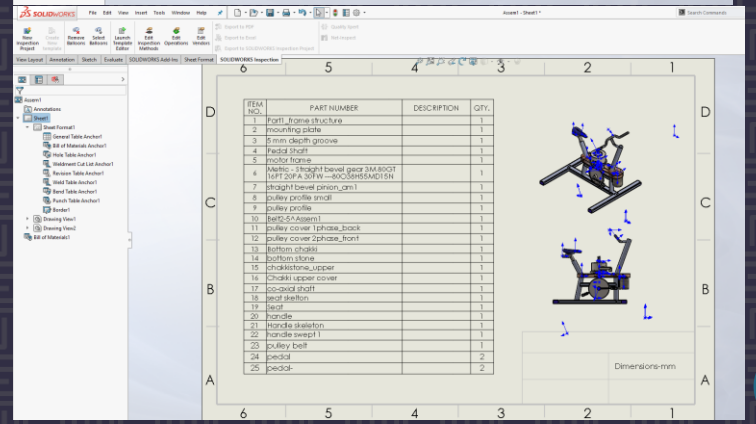


AutoCAD layout Draft



ITEM NO.	QTY.	DESCRIPTION	LENGTH
1	2	TUBE, SQUARE 20.00 X 20.00 X 2.00	320
2	8	TUBE, SQUARE 20.00 X 20.00 X 2.00	125.66
3	4	TUBE, SQUARE 20.00 X 20.00 X 2.00	350
4	2	TUBE, SQUARE 20.00 X 20.00 X 2.00	490
5	1	TUBE, SQUARE 20.00 X 20.00 X 2.00	640.4
6	1	TUBE, SQUARE 20.00 X 20.00 X 2.00	630.63
7	2	TUBE, RECTANGULAR 50.00 X 30.00 X 2.60	460





ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Part1_frame structure		1
2	mounting plate		1
3	5 mm depth groove		1
4	Pedal Shaft		1
5	motor frame		1
6	Metric - Straight bevel gear SM.80GT LEFT.20PA.30FW---80GT80SMD1SN		1
7	straight bevel pinion_am1		1
8	pulley profile small		1
9	pulley profile		1
10	BeltD-5-Assem1		1
11	pulley cover 1phase_back		1
12	pulley cover 2phase_front		1
13	Bottom chakki		1
14	bottom stone		1
15	chakkistone_upper		1
16	Chakki upper cover		1
17	co-pedal shaft		1
18	seat skelton		1
19	seat		1
20	handle		1
21	Handle skeleton		1
22	handle swept 1		1
25	pulley belt		1
24	pedal		2
25	pedal-		2

Solidworks BOM with Draft layout

TOOL DESIGN (INJECTION MOULD)

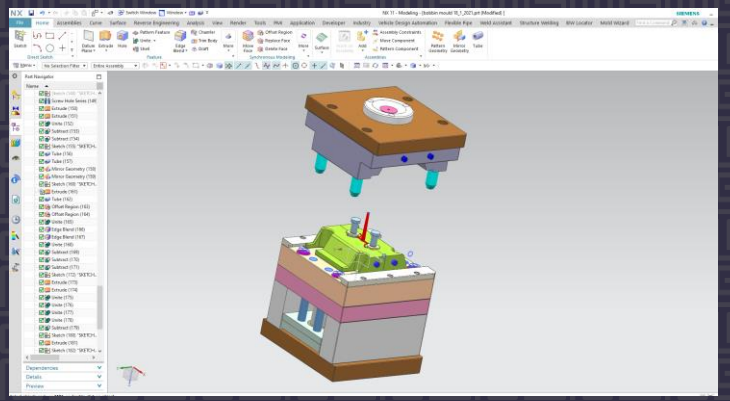
Injection mould tool design:

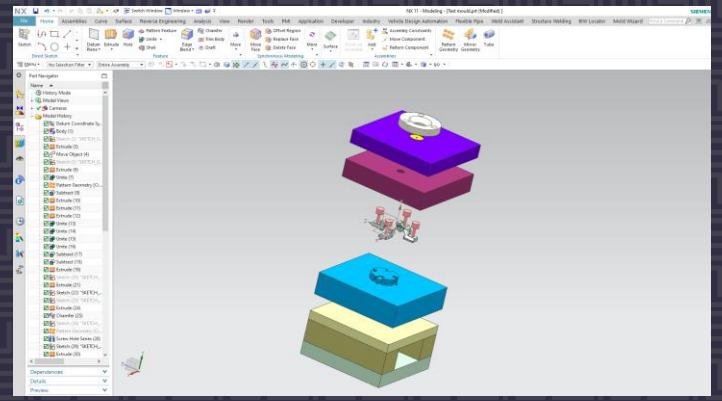
(a) Split mould Tool design

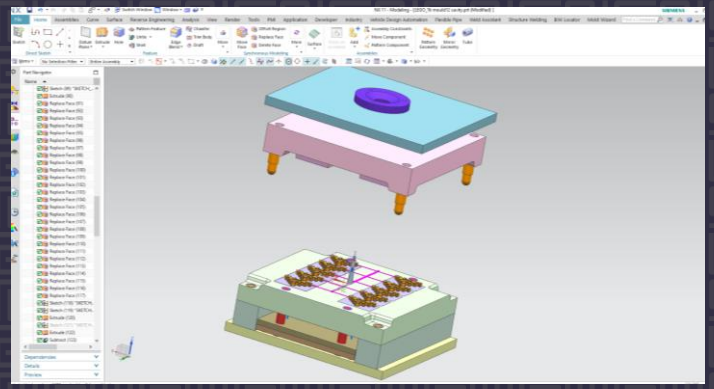
(b) Machine part tool design

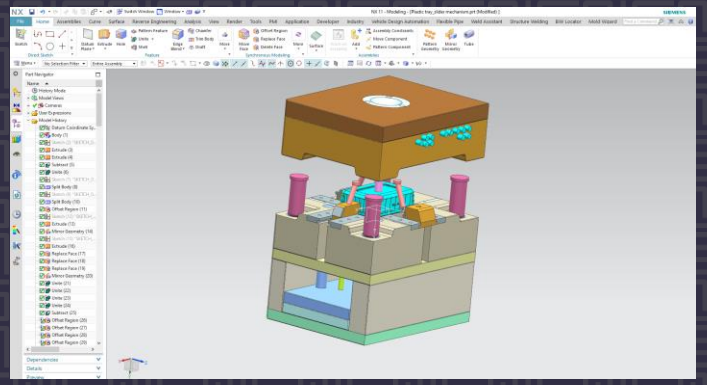
(c) LEGO toy mould design.

(d) Angle pin slider mould basket

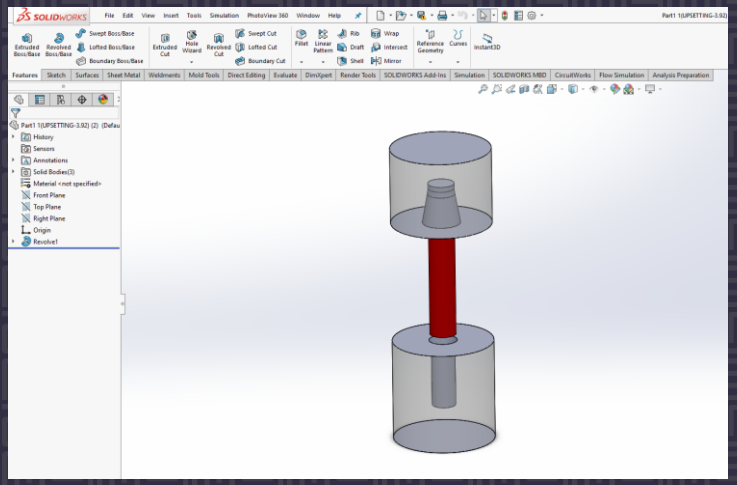


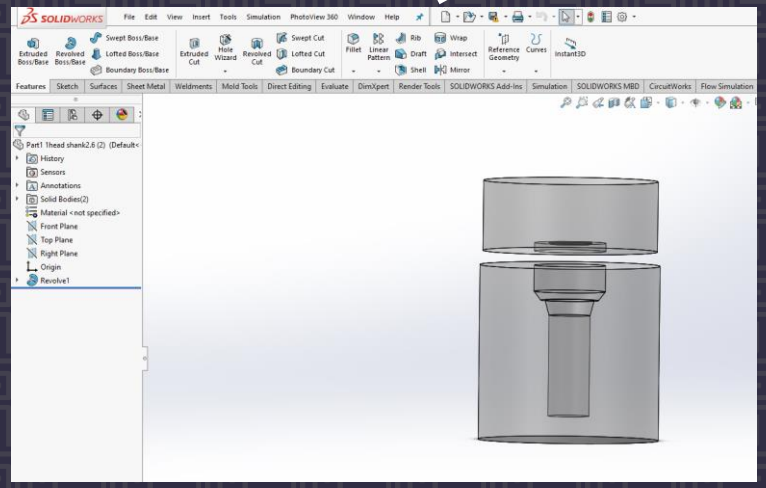


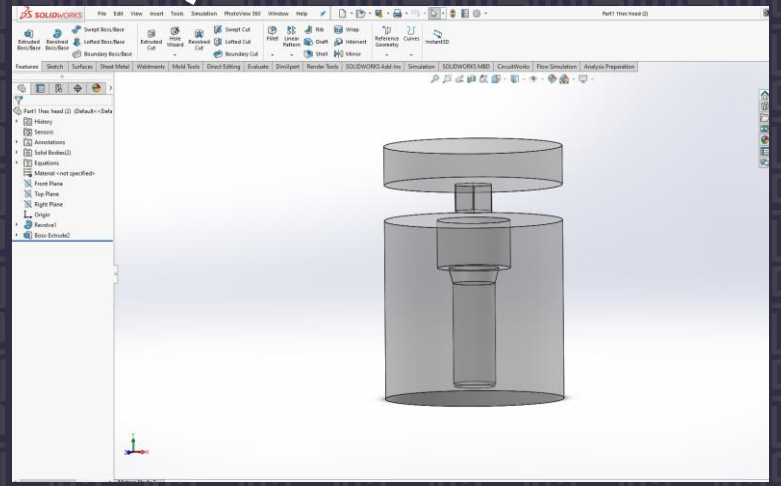




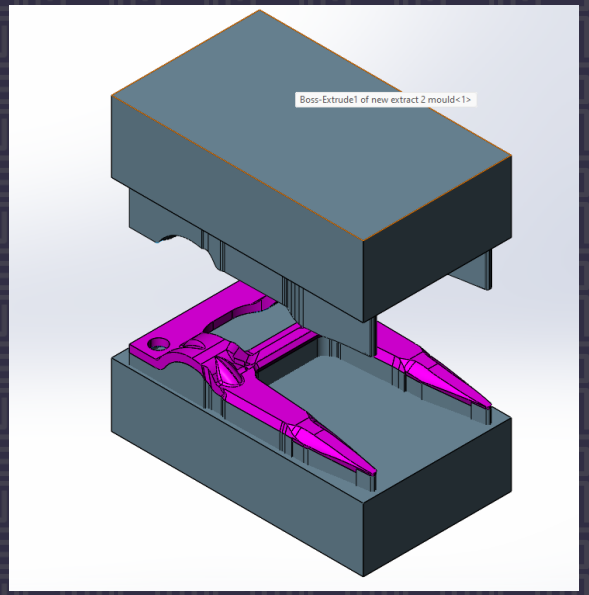
TOOL DESIGN(COLD- FORGE)

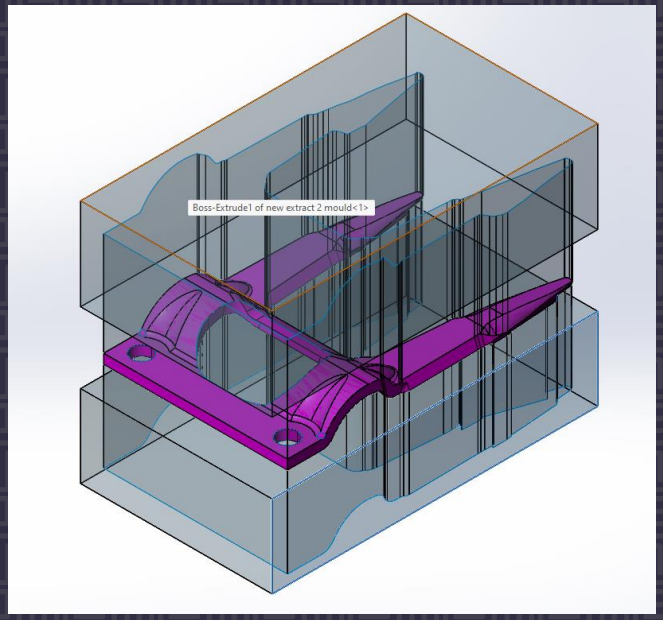






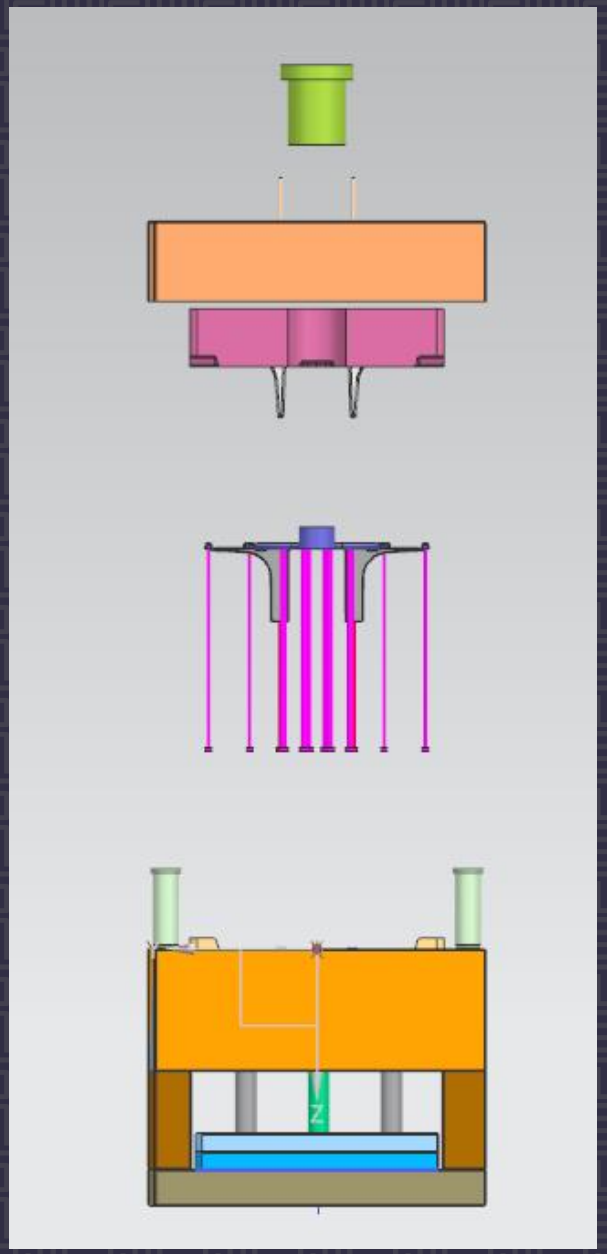
3- Preform of progressive Tool design for cold forming fastener die punch assembly designed in Solidworks.

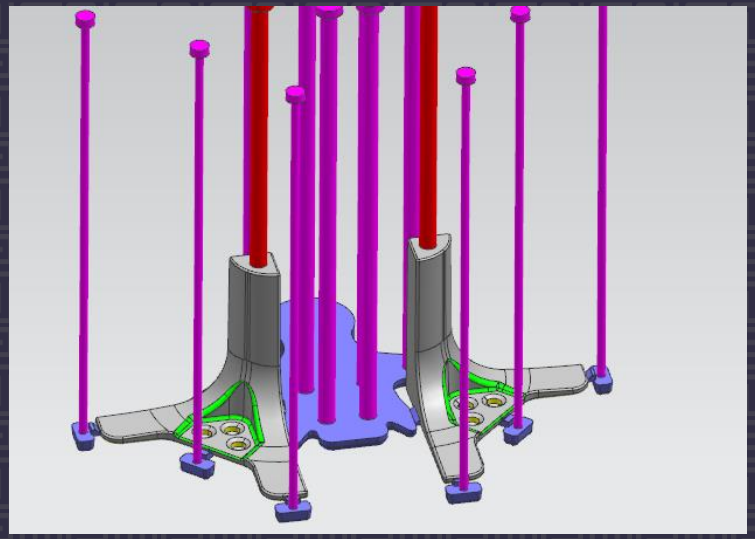


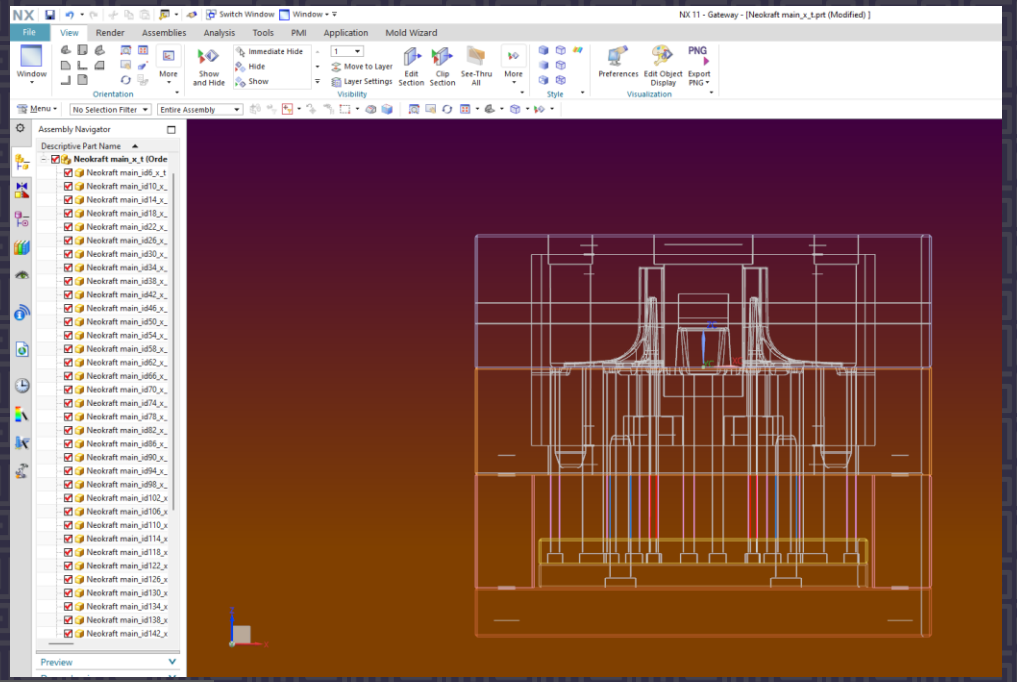


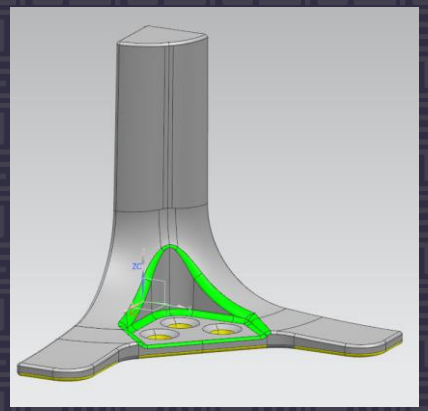
Bending And cutting die punch sheet metal part

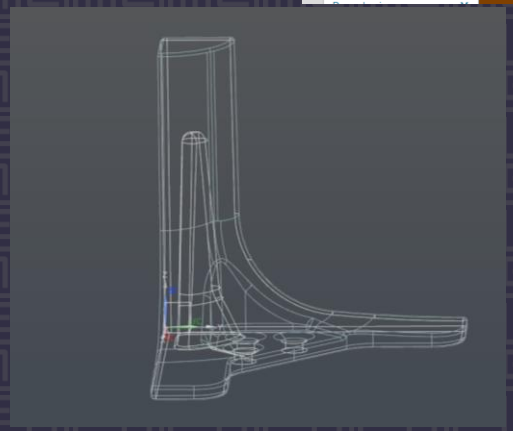
HPDC TOOL DESIGN







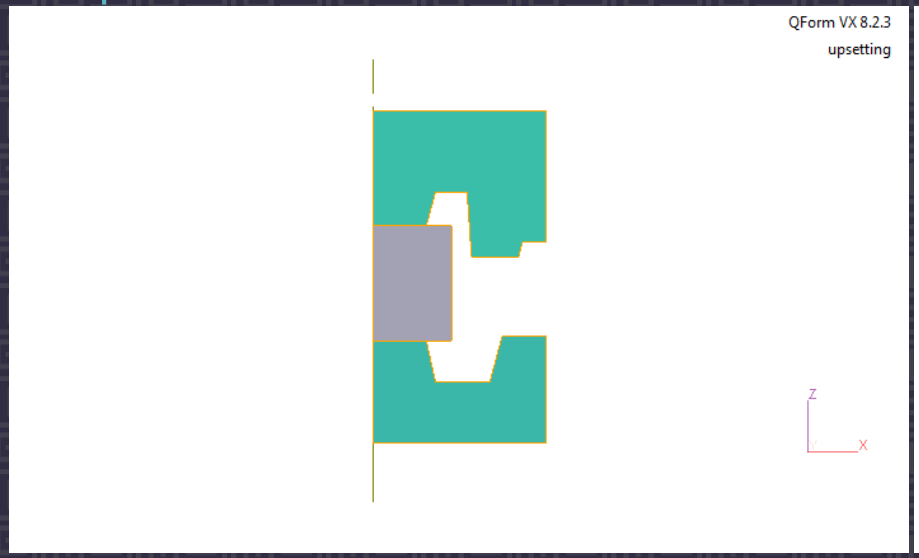


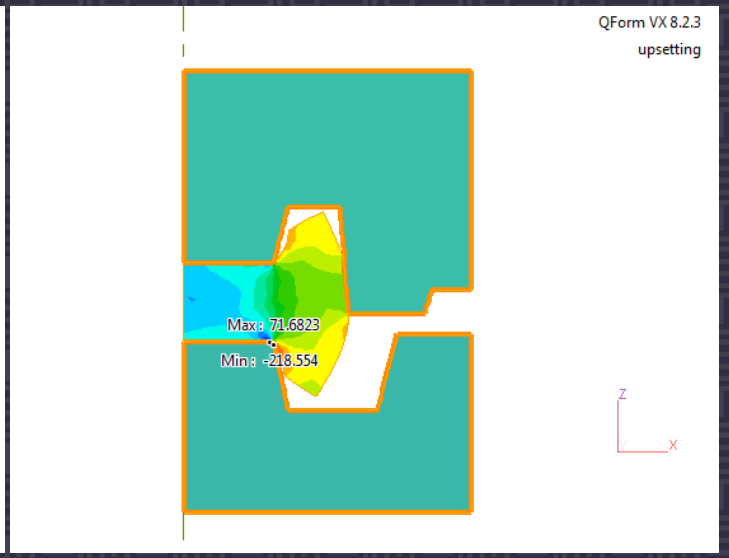


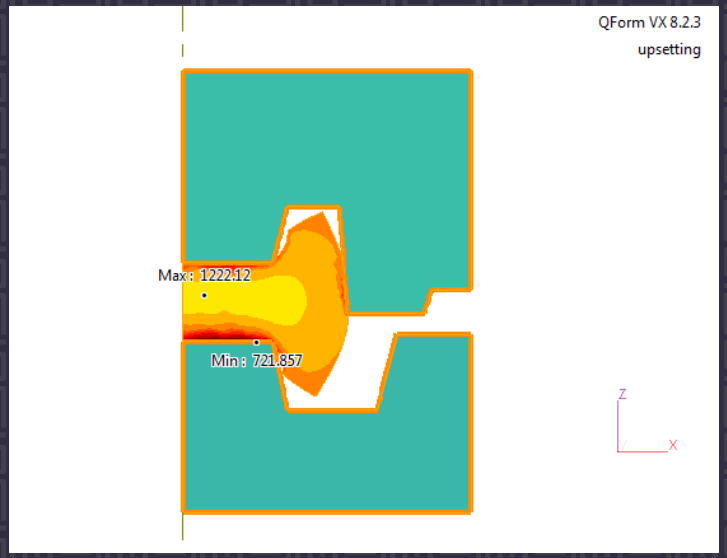
Component

Aluminum part tool designed for 150-180 tonnage HPDC machine, with size of 465*365*550

ANALYSIS-1







- These are 2-D simulation done on Qform. The 2-D simulation for cold forging are done for decreasing the processing time and iterations to reach a conclusion of design in minimum time, neglecting the insignificant results which not contribute higher Percentage on designing of that specific tool.

ANALYSIS-2

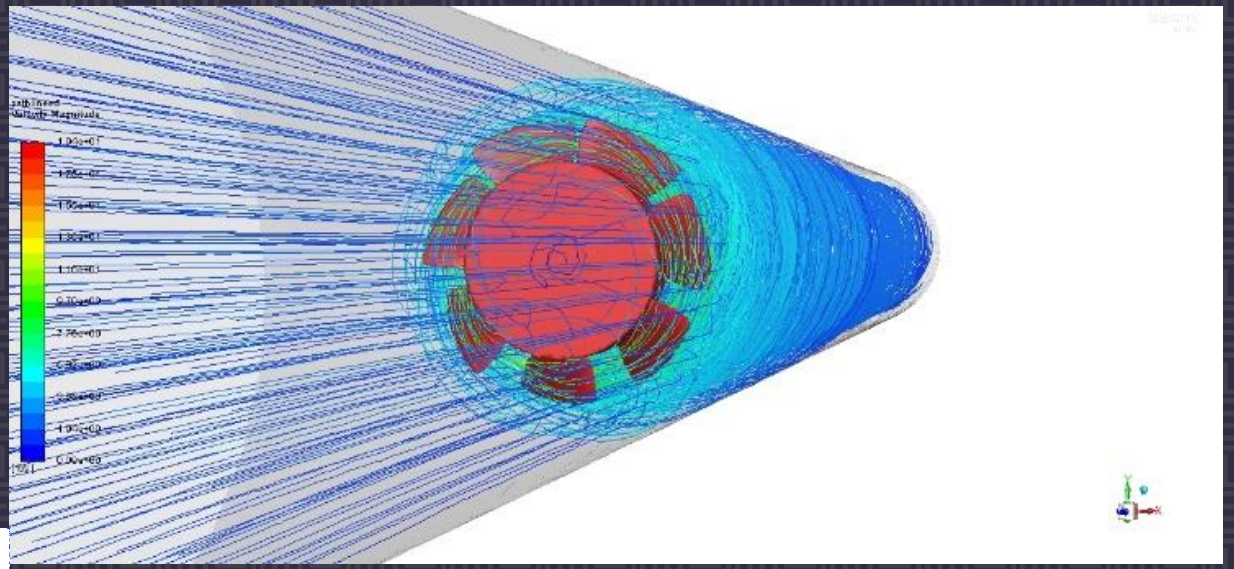
Ansys Workbench- Fluent

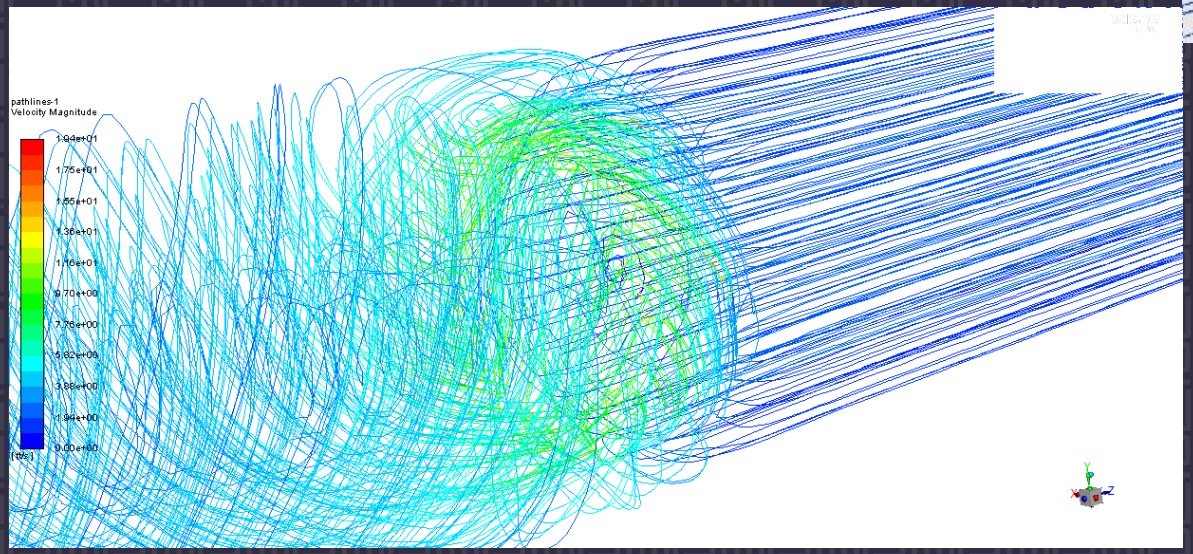
- Analysis of Fan blade with Inlet/Outlet conditions Inside the boundary region.

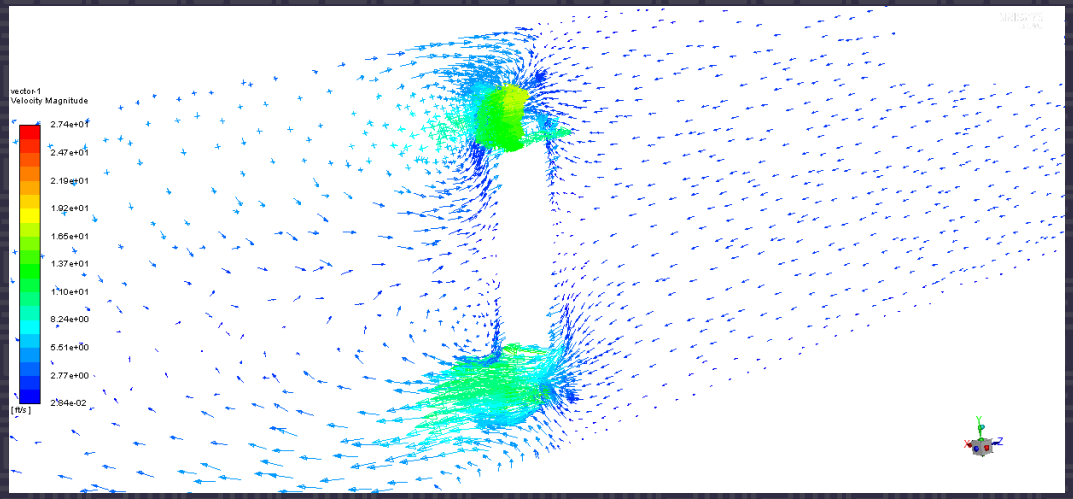
(a) FFT result analysis

(b) Pathlines analysis

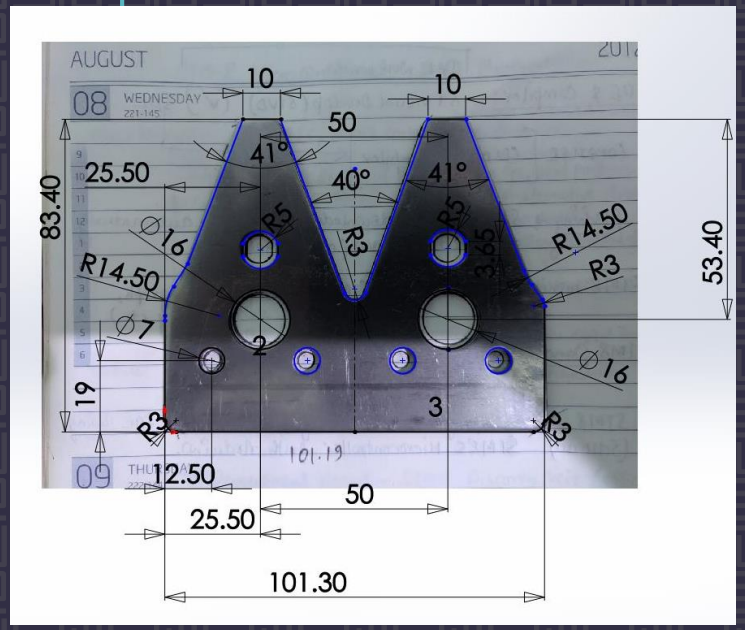
(c) Vector analysis across plane (2-D plane anlysis)

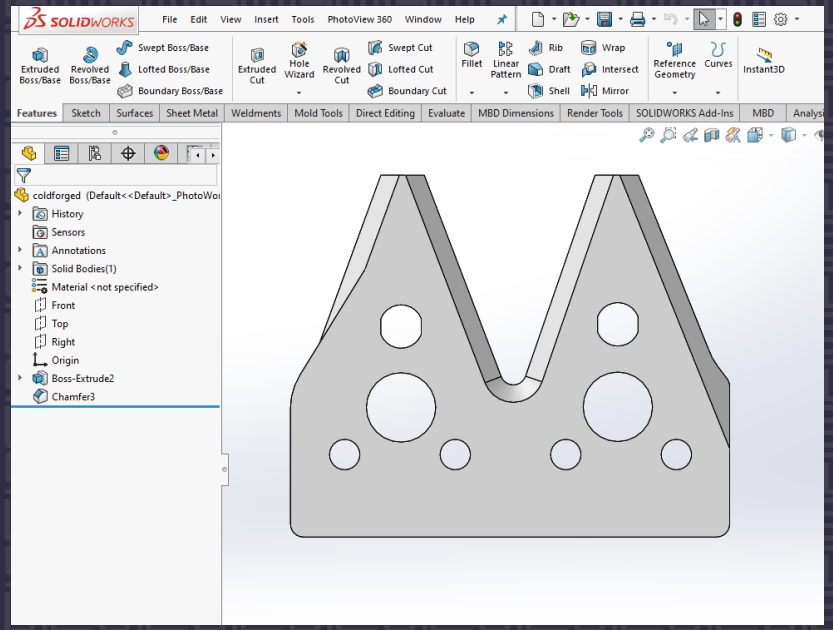






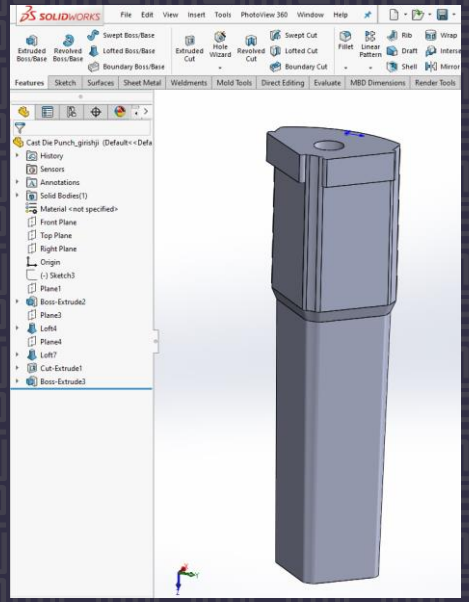
REVERSE ENGINEERING



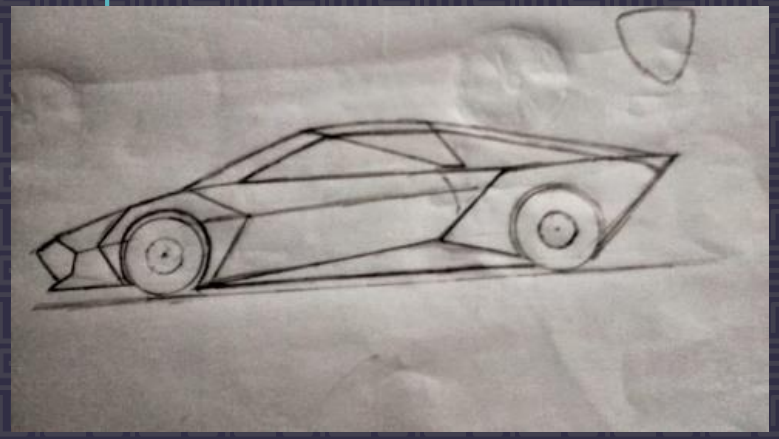




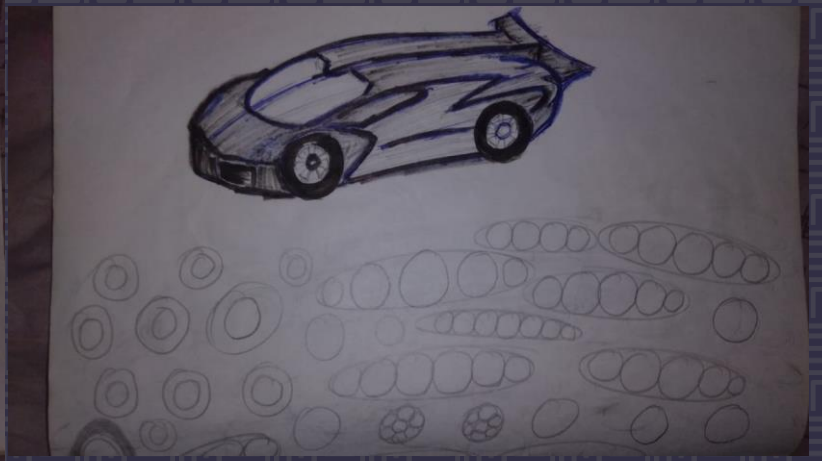




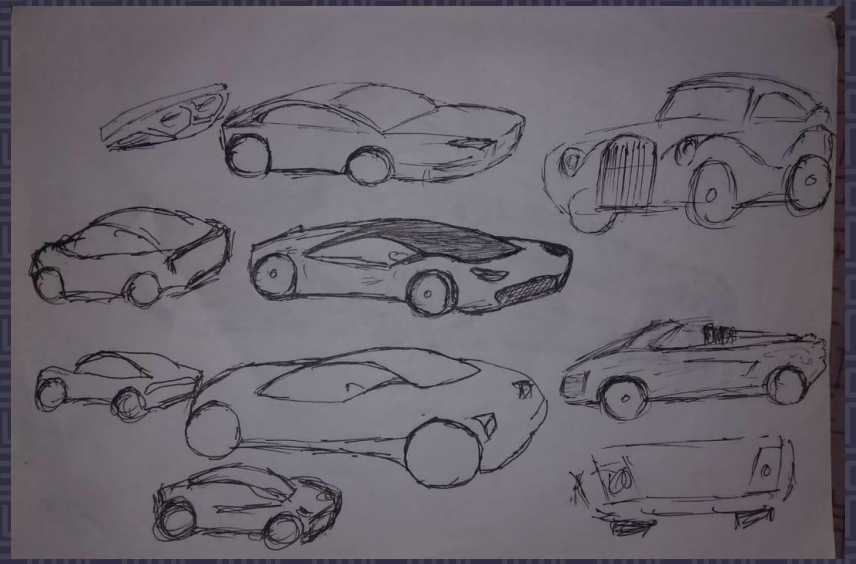
FREE HAND SKETCH-EXTRA



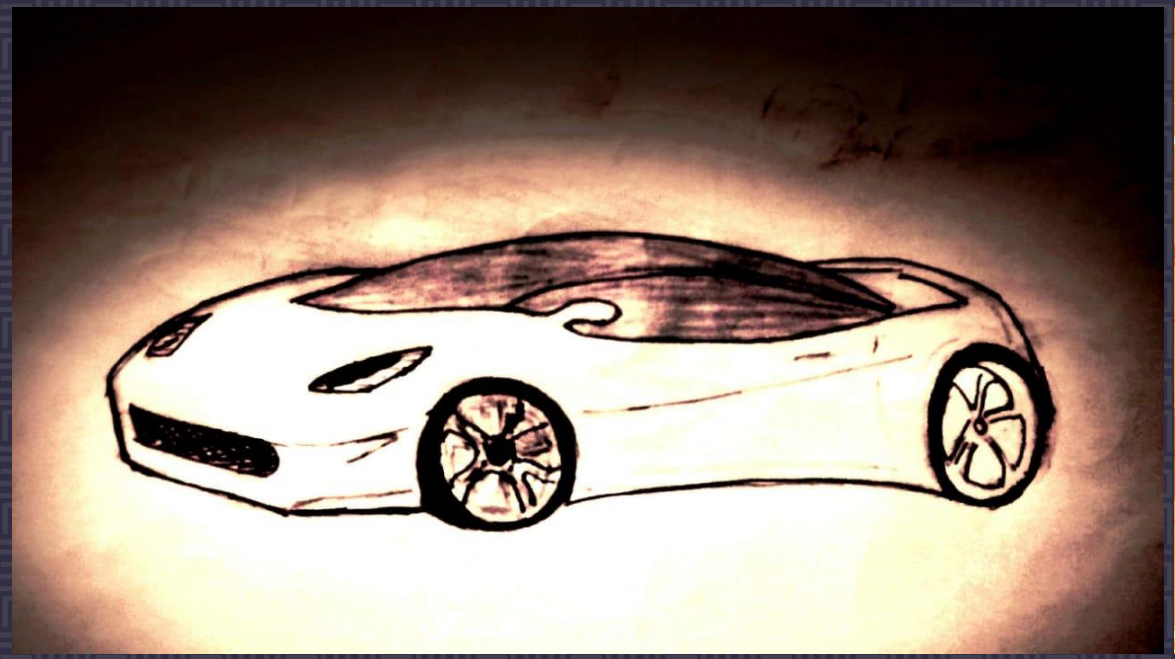


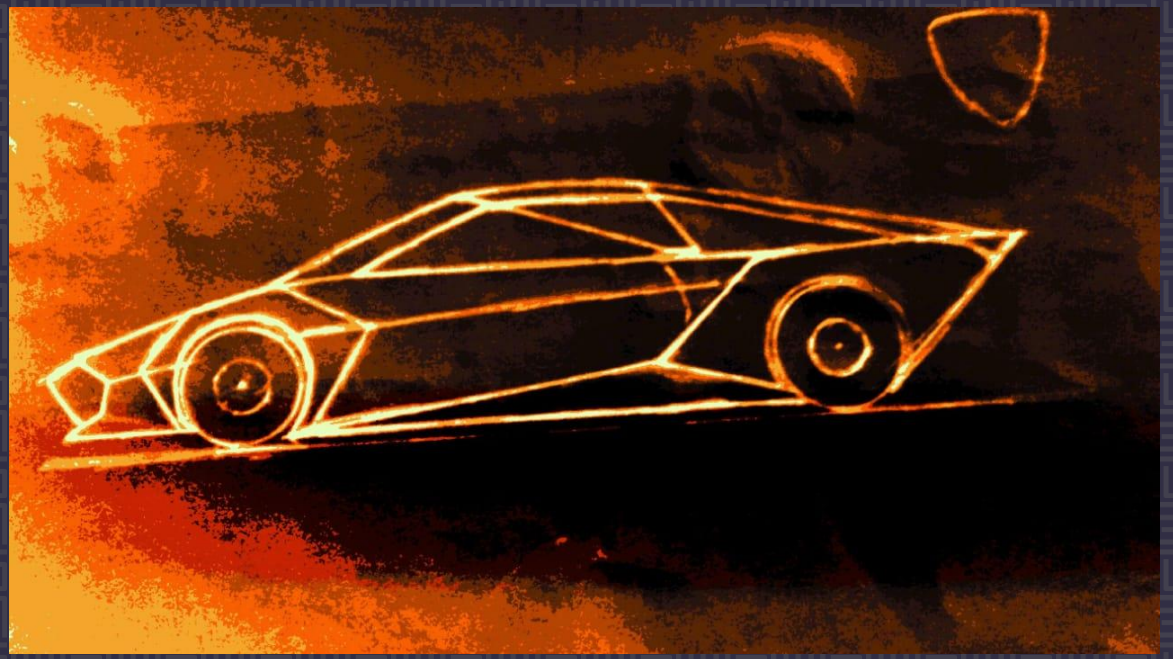






FREE HAND SKETCH & EDITING





Created and edited with Picasa

NOTE

- All the designs of parts, assembly, tools designed are projects done via freelance Intern work and self-learning project during bachelors and master courses (whatever found self proclaimed) not for particular industry project related.

Self- Declaration: All the drawing and parts screenshot taken from my system and made by myself, if any case found or proven copied/not original ready to bear its responsibility.

THANKS